AppealPath

From a confusing health insurance denial letter to the right next step.

DSH Hacks V2 · AI × Healthcare · **Faster care**: gets denied care moving again · built solo by Noah · English and Spanish

Prototype. Not medical or legal advice. All letters shown or tested are fictional; no real patient data was used.

THE PROBLEM

When a health plan denies care, the letter is full of codes, policy names and fine print. People don't know whether to call their doctor's office, file an appeal, or ask for an urgent review, and the deadline is easy to miss. The care the doctor already ordered just waits. A denial is a barrier to *access*, not a diagnosis problem.

WHO IT'S FOR

Patients and caregivers on job-based or Marketplace plans holding a denial letter, often on a phone and stressed.

HOW IT WORKS

1. Take a photo of the letter and answer 5 questions (plan type, has the care happened, does your doctor say waiting could seriously harm you, has the office resubmitted, your state).
2. An AI vision model (Gemma 4 on Cloudflare Workers AI) copies the key facts: what was denied, why, the deadline, who to call. It also suggests the kind of reason.
3. A **second reader** (Tesseract, running on the user's own device) looks for the deadline, reason and phone on the photo, highlights where they're printed, and flags any it can't find, including fax numbers picked as phone numbers.
4. The user checks and corrects the deadline and the reason. Then fixed rules, each with a source or labeled a design choice, pick one next step:

Fix & resubmit	**Appeal**	**Urgent review today**
Missing prior authorization, records or a billing fix	Not medically necessary, not covered, out of network	Care is still ahead and waiting could cause harm

Results give free help (Arkansas contacts), a call script (fix, urgent) or a printable appeal letter filled with the letter's own words (appeal), and a calendar reminder when a date is printed. Medicare, Medicaid and unclear letters get "call the number on your letter" instead of a guess. Five sample letters, one for each rule, open from direct links (`?sample=L01` … `L12`).

WHAT MAKES IT DIFFERENT

- **The AI reads; rules decide.** The user can correct anything it read, including the kind of reason. Each rule cites a source (Healthcare.gov) or is labeled a design choice.
- **It doesn't trust AI confidence.** In testing, the model rated itself 100% sure about a sentence it had copied with a word missing. So a second reader checks its words on the photo, and the user checks before anything is decided.
- **It never makes up a deadline.** It quotes the one printed on the letter, or says there isn't one.
- **Saves no photos or answers:** no accounts, no database; the Worker passes the photo to Workers AI and logs only cost and timing (src/worker.ts).

HOW WELL IT WORKS

TEST IMAGES (FICTIONAL LETTERS)	RIGHT NEXT STEP	SECOND READER (DEADLINE, PHONE)
12 clean digital images	**12 / 12**	21/21 correct values confirmed; 2/2 wrong ones flagged
14 simulated phone photos (tilted, blurred, shadowed, compressed)	**14 / 14**	22/24 confirmed; 2/2 wrong ones flagged

Known misses: the AI sometimes picked a fax number as the phone, and the second reader flagged every one. The reading instructions were tuned on these same letters, so these are best-case numbers; real-world accuracy will be lower, which is why every uploaded letter goes through the check screen.

WHAT IT WON'T DO

Contact insurers, file anything, predict whether an appeal will win, diagnose, give legal advice, or store photos or answers.

BUILT WITH

Cloudflare Workers (free plan) and Workers AI (Gemma 4, Llama 4 Scout as backup); Tesseract.js in the browser; Preact; TypeScript; 45 automated tests. Checked against WCAG 2.1 AA (7 issues found and fixed). AI coding tools were used during the build, as DSH allows.

SOURCES

Healthcare.gov (internal appeals; external review) · Arkansas Insurance Department (consumer services; external review) · U.S. Department of Labor (EBSA)

Try it: appealpath.7zhvt4rjzh.workers.dev · **Code:** github.com/h6ctqtwkkg-byte/appealpath · **Demo video:** add the link here before submitting